

sg13g2_io_fast_1p32V_3p6V_m40C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx



*sg13g2_io_fast_1p32V_3p6V_m40C Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C, Voltage 1.32, Temp
-40.00*

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.03003	0.02756	0.29002	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.03003	0.02756	0.37331	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.03003	0.02756	0.21785	0.24000	12.00000
sg13g2_IOPadOut16mA	0.04180	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.04181	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.04179	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	104.33700	2506.96533	4914.12000
sg13g2_IOPadInOut30mA	212.60500	2807.30606	4914.80000
sg13g2_IOPadInOut4mA	134.48000	2320.90072	4914.19000
sg13g2_IOPadOut16mA	381.59100	900.55700	1334.97000
sg13g2_IOPadOut30mA	592.97600	1197.98833	2170.40000
sg13g2_IOPadOut4mA	257.77400	679.90667	1335.32000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (RR)	0.12000	0.02400	0.17095	0.80000	0.14400	0.30840	5.00000	0.24000	0.52879
sg13g2_IOPadInOut30mA	pad->p2c (RR)	0.12000	0.02400	0.17115	0.80000	0.14400	0.30791	5.00000	0.24000	0.52911
sg13g2_IOPadInOut4mA	pad->p2c (RR)	0.12000	0.02400	0.17097	0.80000	0.14400	0.30828	5.00000	0.24000	0.52870

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (FF)	0.12000	0.02400	0.16989	0.80000	0.14400	0.31519	5.00000	0.24000	0.57829
sg13g2_IOPadInOut30mA	pad->p2c (FF)	0.12000	0.02400	0.16993	0.80000	0.14400	0.31537	5.00000	0.24000	0.57844
sg13g2_IOPadInOut4mA	pad->p2c (FF)	0.12000	0.02400	0.16961	0.80000	0.14400	0.31495	5.00000	0.24000	0.57600

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (RR)	0.02000	1.00000	1.08330	0.33000	9.00000	1.81966	2.50000	20.00000	2.71372
	c2p_en->pad (FR)	0.02000	1.00000	0.79693	0.33000	9.00000	0.84963	2.50000	20.00000	0.99312
	c2p_en->pad (RR)	0.02000	1.00000	1.07760	0.33000	9.00000	1.85144	2.50000	20.00000	2.80298
sg13g2_IOPadInOut30mA	c2p->pad (RR)	0.02000	2.00000	1.35361	0.33000	18.00000	2.25832	2.50000	40.00000	3.22849
	c2p_en->pad (FR)	0.02000	2.00000	0.95564	0.33000	18.00000	1.00974	2.50000	40.00000	1.15442
	c2p_en->pad (RR)	0.02000	2.00000	1.33796	0.33000	18.00000	2.29398	2.50000	40.00000	3.31981
sg13g2_IOPadInOut4mA	c2p->pad (RR)	0.02000	1.00000	1.07469	0.33000	5.00000	2.32568	2.50000	12.00000	4.49301
	c2p_en->pad (FR)	0.02000	1.00000	0.63989	0.33000	5.00000	0.69237	2.50000	12.00000	0.83906
	c2p_en->pad (RR)	0.02000	1.00000	1.07785	0.33000	5.00000	2.37631	2.50000	12.00000	4.63825
sg13g2_IOPadOut16mA	c2p->pad (RR)	0.02000	1.00000	1.03402	0.33000	9.00000	1.85418	2.50000	20.00000	3.47838
sg13g2_IOPadOut30mA	c2p->pad (RR)	0.02000	2.00000	1.30659	0.33000	18.00000	2.29340	2.50000	40.00000	3.99769
sg13g2_IOPadOut4mA	c2p->pad (RR)	0.02000	1.00000	1.01770	0.33000	5.00000	2.35253	2.50000	12.00000	5.24966

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	0.86216	0.33000	9.00000	1.53629	2.50000	20.00000	2.59898
	c2p_en->pad (FF)	0.02000	1.00000	0.85643	0.33000	9.00000	0.92118	2.50000	20.00000	1.13378
	c2p_en->pad (RF)	0.02000	1.00000	0.73512	0.33000	9.00000	1.48753	2.50000	20.00000	2.45199
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	1.12613	0.33000	18.00000	1.88509	2.50000	40.00000	2.97448
	c2p_en->pad (FF)	0.02000	2.00000	1.27839	0.33000	18.00000	1.35061	2.50000	40.00000	1.54213
	c2p_en->pad (RF)	0.02000	2.00000	0.85347	0.33000	18.00000	1.71331	2.50000	40.00000	2.74322
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	0.90776	0.33000	5.00000	2.19820	2.50000	12.00000	4.59233
	c2p_en->pad (FF)	0.02000	1.00000	0.49359	0.33000	5.00000	0.56320	2.50000	12.00000	0.77177
	c2p_en->pad (RF)	0.02000	1.00000	0.92321	0.33000	5.00000	2.30224	2.50000	12.00000	4.68541
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	0.80338	0.33000	9.00000	1.44653	2.50000	20.00000	2.36105
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	1.07000	0.33000	18.00000	1.80025	2.50000	40.00000	2.73951
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	0.83148	0.33000	5.00000	2.09453	2.50000	12.00000	4.33705

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	-1.42309	0.80000	0.14400	-1.16828	5.00000	0.24000	0.52348
	pad	0.12000	0.02400	0.03651	0.80000	0.14400	0.03860	5.00000	0.24000	0.05056
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	-2.24215	0.80000	0.14400	-1.97655	5.00000	0.24000	-0.28715
	pad	0.12000	0.02400	0.03656	0.80000	0.14400	0.03865	5.00000	0.24000	0.05072
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	-0.76363	0.80000	0.14400	-0.51039	5.00000	0.24000	1.17858
	pad	0.12000	0.02400	0.03630	0.80000	0.14400	0.03848	5.00000	0.24000	0.05042

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	2.02308	0.80000	0.14400	2.28831	5.00000	0.24000	3.99929
	pad	0.12000	0.02400	0.09484	0.80000	0.14400	0.11350	5.00000	0.24000	0.17994
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	2.82743	0.80000	0.14400	3.10104	5.00000	0.24000	4.81689
	pad	0.12000	0.02400	0.09446	0.80000	0.14400	0.11319	5.00000	0.24000	0.17966
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	1.37072	0.80000	0.14400	1.63115	5.00000	0.24000	3.33563
	pad	0.12000	0.02400	0.09215	0.80000	0.14400	0.11071	5.00000	0.24000	0.17863

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	8.17346	0.33000	9.00000	8.08636	2.50000	20.00000	7.73025
	c2p	0.02000	1.00000	1.81339	0.33000	9.00000	15.57460	2.50000	20.00000	22.30700
	c2p_en	0.02000	1.00000	7.60676	0.33000	9.00000	10.36370	2.50000	20.00000	14.81580
	c2p_en	0.02000	1.00000	1.77595	0.33000	9.00000	15.54830	2.50000	20.00000	22.23250
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	13.02880	0.33000	18.00000	11.96740	2.50000	40.00000	10.89180
	c2p	0.02000	2.00000	3.53070	0.33000	18.00000	28.38630	2.50000	40.00000	5.38293
	c2p_en	0.02000	2.00000	11.65620	0.33000	18.00000	16.63430	2.50000	40.00000	22.35530
	c2p_en	0.02000	2.00000	3.49556	0.33000	18.00000	28.39610	2.50000	40.00000	5.22698
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	4.66449	0.33000	5.00000	5.22635	2.50000	12.00000	6.22308
	c2p	0.02000	1.00000	1.82325	0.33000	5.00000	8.71298	2.50000	12.00000	20.96190
	c2p_en	0.02000	1.00000	4.34902	0.33000	5.00000	5.90094	2.50000	12.00000	8.60891
	c2p_en	0.02000	1.00000	1.78551	0.33000	5.00000	8.68364	2.50000	12.00000	20.94860
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	7.70822	0.33000	9.00000	7.39048	2.50000	20.00000	7.23542
	c2p	0.02000	1.00000	-0.02316	0.33000	9.00000	-0.00662	2.50000	20.00000	0.17897
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	12.72980	0.33000	18.00000	11.61830	2.50000	40.00000	11.25140
	c2p	0.02000	2.00000	-0.02316	0.33000	18.00000	-0.00662	2.50000	40.00000	0.17905
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	3.90290	0.33000	5.00000	3.80509	2.50000	12.00000	3.88361
	c2p	0.02000	1.00000	-0.02316	0.33000	5.00000	-0.00662	2.50000	12.00000	0.17844

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	18.96360	0.33000	9.00000	12.60980	2.50000	20.00000	10.58190
	c2p	0.02000	1.00000	0.24990	0.33000	9.00000	0.28231	2.50000	20.00000	0.49082
	c2p_en	0.02000	1.00000	4.36555	0.33000	9.00000	4.75955	2.50000	20.00000	5.35607
	c2p_en	0.02000	1.00000	0.14028	0.33000	9.00000	0.17156	2.50000	20.00000	0.29835
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	68.80180	0.33000	18.00000	42.35730	2.50000	40.00000	31.12050
	c2p	0.02000	2.00000	0.25417	0.33000	18.00000	0.28328	2.50000	40.00000	0.48346
	c2p_en	0.02000	2.00000	6.58565	0.33000	18.00000	7.00550	2.50000	40.00000	7.63739
	c2p_en	0.02000	2.00000	0.14140	0.33000	18.00000	0.17231	2.50000	40.00000	0.29180
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	3.50098	0.33000	5.00000	4.35563	2.50000	12.00000	5.88796
	c2p	0.02000	1.00000	0.26087	0.33000	5.00000	0.30962	2.50000	12.00000	0.54401
	c2p_en	0.02000	1.00000	2.68596	0.33000	5.00000	3.55117	2.50000	12.00000	5.06606
	c2p_en	0.02000	1.00000	0.15188	0.33000	5.00000	0.19857	2.50000	12.00000	0.35326
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	20.20840	0.33000	9.00000	12.25890	2.50000	20.00000	9.76976
	c2p	0.02000	1.00000	0.07638	0.33000	9.00000	0.09817	2.50000	20.00000	0.28176
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	71.89170	0.33000	18.00000	42.87330	2.50000	40.00000	31.51470
	c2p	0.02000	2.00000	0.07637	0.33000	18.00000	0.09816	2.50000	40.00000	0.28258
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.95197	0.33000	5.00000	2.84005	2.50000	12.00000	2.81579
	c2p	0.02000	1.00000	0.07638	0.33000	5.00000	0.09818	2.50000	12.00000	0.28684

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02360	0.33000	-0.02448	2.50000	-0.02494
sg13g2_IOPadInOut30mA	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02360	0.33000	-0.02449	2.50000	-0.02494
sg13g2_IOPadInOut4mA	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02360	0.33000	-0.02453	2.50000	-0.02494

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03062	0.33000	0.03047	2.50000	0.02992
sg13g2_IOPadInOut30mA	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03063	0.33000	0.03046	2.50000	0.02992
sg13g2_IOPadInOut4mA	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03064	0.33000	0.03050	2.50000	0.02991

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	-0.00008	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02360	0.33000	-0.02449	2.50000	-0.02494
	(!c2p_en * !pad * !p2c)	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02360	0.33000	-0.02448	2.50000	-0.02494
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02360	0.33000	-0.02450	2.50000	-0.02494
	(!c2p_en * !pad * !p2c)	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02360	0.33000	-0.02449	2.50000	-0.02494
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02360	0.33000	-0.02454	2.50000	-0.02494
	(!c2p_en * !pad * !p2c)	0.02000	-0.00009	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02360	0.33000	-0.02453	2.50000	-0.02494

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	0.00008	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03063	0.33000	0.03047	2.50000	0.02992
	(!c2p_en * !pad * !p2c)	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03062	0.33000	0.03047	2.50000	0.02992
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03063	0.33000	0.03045	2.50000	0.02992
	(!c2p_en * !pad * !p2c)	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03063	0.33000	0.03046	2.50000	0.02992
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03064	0.33000	0.03050	2.50000	0.02991
	(!c2p_en * !pad * !p2c)	0.02000	0.00009	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03064	0.33000	0.03050	2.50000	0.02991

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	2.01707	0.33000	2.01758	2.50000	2.01946
	0.02000	0.03874	0.33000	0.04631	2.50000	0.15479
sg13g2_IOPadInOut30mA	0.02000	3.21605	0.33000	3.21633	2.50000	3.23100
	0.02000	0.03873	0.33000	0.04632	2.50000	0.15479
sg13g2_IOPadInOut4mA	0.02000	0.99969	0.33000	0.99987	2.50000	1.00099
	0.02000	0.03873	0.33000	0.04635	2.50000	0.15481

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.44023	0.33000	0.43806	2.50000	0.43601
	0.02000	0.10607	0.33000	0.11517	2.50000	0.22385
sg13g2_IOPadInOut30mA	0.02000	0.42625	0.33000	0.42325	2.50000	0.42242
	0.02000	0.10608	0.33000	0.11513	2.50000	0.22377
sg13g2_IOPadInOut4mA	0.02000	0.47587	0.33000	0.47327	2.50000	0.47149
	0.02000	0.10609	0.33000	0.11513	2.50000	0.22395

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	8.17346	0.33000	8.08636	2.50000	7.73025
	0.02000	1.81339	0.33000	15.57460	2.50000	22.30700
	0.02000	7.60676	0.33000	10.36370	2.50000	14.81580
	0.02000	1.77595	0.33000	15.54830	2.50000	22.23250
sg13g2_IOPadInOut30mA	0.02000	13.02880	0.33000	11.96740	2.50000	10.89180
	0.02000	3.53070	0.33000	28.38630	2.50000	5.38293
	0.02000	11.65620	0.33000	16.63430	2.50000	22.35530
	0.02000	3.49556	0.33000	28.39610	2.50000	5.22698
sg13g2_IOPadInOut4mA	0.02000	4.66449	0.33000	5.22635	2.50000	6.22308
	0.02000	1.82325	0.33000	8.71298	2.50000	20.96190
	0.02000	4.34902	0.33000	5.90094	2.50000	8.60891
	0.02000	1.78551	0.33000	8.68364	2.50000	20.94860
sg13g2_IOPadOut16mA	0.02000	7.70822	0.33000	7.39048	2.50000	7.23542
	0.02000	-0.02316	0.33000	-0.00662	2.50000	0.17897
sg13g2_IOPadOut30mA	0.02000	12.72980	0.33000	11.61830	2.50000	11.25140
	0.02000	-0.02316	0.33000	-0.00662	2.50000	0.17905
sg13g2_IOPadOut4mA	0.02000	3.90290	0.33000	3.80509	2.50000	3.88361
	0.02000	-0.02316	0.33000	-0.00662	2.50000	0.17844

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	18.96360	0.33000	12.60980	2.50000	10.58190
	0.02000	0.24990	0.33000	0.28231	2.50000	0.49082
	0.02000	4.36555	0.33000	4.75955	2.50000	5.35607
	0.02000	0.14028	0.33000	0.17156	2.50000	0.29835
sg13g2_IOPadInOut30mA	0.02000	68.80180	0.33000	42.35730	2.50000	31.12050
	0.02000	0.25417	0.33000	0.28328	2.50000	0.48346
	0.02000	6.58565	0.33000	7.00550	2.50000	7.63739
	0.02000	0.14140	0.33000	0.17231	2.50000	0.29180
sg13g2_IOPadInOut4mA	0.02000	3.50098	0.33000	4.35563	2.50000	5.88796
	0.02000	0.26087	0.33000	0.30962	2.50000	0.54401
	0.02000	2.68596	0.33000	3.55117	2.50000	5.06606
	0.02000	0.15188	0.33000	0.19857	2.50000	0.35326
sg13g2_IOPadOut16mA	0.02000	20.20840	0.33000	12.25890	2.50000	9.76976
	0.02000	0.07638	0.33000	0.09817	2.50000	0.28176
sg13g2_IOPadOut30mA	0.02000	71.89170	0.33000	42.87330	2.50000	31.51470
	0.02000	0.07637	0.33000	0.09816	2.50000	0.28258
sg13g2_IOPadOut4mA	0.02000	2.95197	0.33000	2.84005	2.50000	2.81579
	0.02000	0.07638	0.33000	0.09818	2.50000	0.28684

INPUT



*sg13g2_io_fast_1p32V_3p6V_m40C Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C, Voltage 1.32, Temp -40.00*

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23272	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	153.25400	333.17867	673.25800

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (RR)	0.12000	0.02400	0.17085	0.80000	0.14400	0.30734	5.00000	0.24000	0.52778

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (FF)	0.12000	0.02400	0.16939	0.80000	0.14400	0.31477	5.00000	0.24000	0.57485

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	-0.67154	0.80000	0.14400	-0.41892	5.00000	0.24000	1.26499
	pad	0.12000	0.02400	0.03701	0.80000	0.14400	0.03904	5.00000	0.24000	0.05127

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	1.27943	0.80000	0.14400	1.53928	5.00000	0.24000	3.25547
	pad	0.12000	0.02400	0.09116	0.80000	0.14400	0.10970	5.00000	0.24000	0.17788

SG13G2_CORNER



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000



sg13g2_io_fast_1p32V_3p6V_m40C

Cell Library: Process

sg13g2_io_fast_1p32V_3p6V_m40C,

Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG



sg13g2_io_fast_1p32V_3p6V_m40C

Cell Library: Process

sg13g2_io_fast_1p32V_3p6V_m40C,

Voltage 1.32, Temp -40.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.74927	0.61118	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	1359.77000	2719.54000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (R)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (F)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

SG13G2_IOPADIOVDD



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS



sg13g2_io_fast_1p32V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p32V_3p6V_m40C,
Voltage 1.32, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx



*sg13g2_io_fast_1p32V_3p6V_m40C Cell Library:
Process sg13g2_io_fast_1p32V_3p6V_m40C, Voltage
1.32, Temp -40.00*

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.02953	0.02810	20.00000
sg13g2_IOPadTriOut30mA	0.02953	0.02809	40.00000
sg13g2_IOPadTriOut4mA	0.02953	0.02810	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	190.68000	2129.54290	4386.69000
sg13g2_IOPadTriOut30mA	336.93300	2313.49000	4388.41000
sg13g2_IOPadTriOut4mA	81.66150	1980.43651	4386.86000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	1.08256	0.33000	9.00000	1.82045	2.50000	20.00000	2.71205
	c2p_en->pad (FR)	0.02000	1.00000	0.80367	0.33000	9.00000	0.85542	2.50000	20.00000	1.00101
	c2p_en->pad (RR)	0.02000	1.00000	1.07879	0.33000	9.00000	1.85448	2.50000	20.00000	2.80615
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	1.35488	0.33000	18.00000	2.25998	2.50000	40.00000	3.22996
	c2p_en->pad (FR)	0.02000	2.00000	0.96518	0.33000	18.00000	1.01454	2.50000	40.00000	1.16141
	c2p_en->pad (RR)	0.02000	2.00000	1.34016	0.33000	18.00000	2.30225	2.50000	40.00000	3.32554
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	1.06794	0.33000	5.00000	2.31810	2.50000	12.00000	4.48273
	c2p_en->pad (FR)	0.02000	1.00000	0.64950	0.33000	5.00000	0.70203	2.50000	12.00000	0.84653
	c2p_en->pad (RR)	0.02000	1.00000	1.07428	0.33000	5.00000	2.37535	2.50000	12.00000	4.63860

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	0.86512	0.33000	9.00000	1.53920	2.50000	20.00000	2.60376
	c2p_en->pad (FF)	0.02000	1.00000	0.85557	0.33000	9.00000	0.92689	2.50000	20.00000	1.14322
	c2p_en->pad (RF)	0.02000	1.00000	0.73650	0.33000	9.00000	1.48758	2.50000	20.00000	2.44719
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	1.13093	0.33000	18.00000	1.89121	2.50000	40.00000	2.97945
	c2p_en->pad (FF)	0.02000	2.00000	1.28489	0.33000	18.00000	1.35728	2.50000	40.00000	1.55111
	c2p_en->pad (RF)	0.02000	2.00000	0.85671	0.33000	18.00000	1.71894	2.50000	40.00000	2.75166
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	0.90305	0.33000	5.00000	2.19349	2.50000	12.00000	4.58702
	c2p_en->pad (FF)	0.02000	1.00000	0.50033	0.33000	5.00000	0.56939	2.50000	12.00000	0.77886
	c2p_en->pad (RF)	0.02000	1.00000	0.92079	0.33000	5.00000	2.30232	2.50000	12.00000	4.69105

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	7.62292	0.33000	9.00000	7.36160	2.50000	20.00000	6.82438
	c2p	0.02000	1.00000	0.06348	0.33000	9.00000	0.07164	2.50000	20.00000	0.25755
	c2p_en	0.02000	1.00000	6.98638	0.33000	9.00000	9.63947	2.50000	20.00000	12.96320
	c2p_en	0.02000	1.00000	0.02562	0.33000	9.00000	0.04092	2.50000	20.00000	0.24193
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	12.57730	0.33000	18.00000	11.61820	2.50000	40.00000	11.10310
	c2p	0.02000	2.00000	0.06349	0.33000	18.00000	0.07165	2.50000	40.00000	0.25771
	c2p_en	0.02000	2.00000	11.07540	0.33000	18.00000	16.92120	2.50000	40.00000	22.37220
	c2p_en	0.02000	2.00000	0.02563	0.33000	18.00000	0.04093	2.50000	40.00000	0.24682
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	3.89678	0.33000	5.00000	3.74850	2.50000	12.00000	3.52224
	c2p	0.02000	1.00000	0.06350	0.33000	5.00000	0.07166	2.50000	12.00000	0.25744
	c2p_en	0.02000	1.00000	3.58457	0.33000	5.00000	4.46447	2.50000	12.00000	6.08094
	c2p_en	0.02000	1.00000	0.02565	0.33000	5.00000	0.04094	2.50000	12.00000	0.24475

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	18.44140	0.33000	9.00000	11.71030	2.50000	20.00000	9.10548
	c2p	0.02000	1.00000	0.15072	0.33000	9.00000	0.16161	2.50000	20.00000	0.34929
	c2p_en	0.02000	1.00000	3.85629	0.33000	9.00000	3.85883	2.50000	20.00000	3.85702
	c2p_en	0.02000	1.00000	0.04240	0.33000	9.00000	0.05034	2.50000	20.00000	0.15610
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	68.01810	0.33000	18.00000	41.47560	2.50000	40.00000	29.51590
	c2p	0.02000	2.00000	0.15074	0.33000	18.00000	0.16163	2.50000	40.00000	0.34935
	c2p_en	0.02000	2.00000	6.06982	0.33000	18.00000	6.07425	2.50000	40.00000	6.07769
	c2p_en	0.02000	2.00000	0.04275	0.33000	18.00000	0.05037	2.50000	40.00000	0.15607
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.85974	0.33000	5.00000	2.84636	2.50000	12.00000	2.85982
	c2p	0.02000	1.00000	0.15087	0.33000	5.00000	0.16176	2.50000	12.00000	0.34914
	c2p_en	0.02000	1.00000	2.02368	0.33000	5.00000	2.02135	2.50000	12.00000	2.02109
	c2p_en	0.02000	1.00000	0.04285	0.33000	5.00000	0.05051	2.50000	12.00000	0.15765

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	-0.00005	0.33000	-0.00001	2.50000	-0.00002
	0.02000	-0.02369	0.33000	-0.02467	2.50000	-0.02522
sg13g2_IOPadTriOut30mA	0.02000	0.00004	0.33000	0.00008	2.50000	0.00007
	0.02000	-0.02369	0.33000	-0.02467	2.50000	-0.02522
sg13g2_IOPadTriOut4mA	0.02000	-0.00007	0.33000	-0.00006	2.50000	-0.00006
	0.02000	-0.02370	0.33000	-0.02468	2.50000	-0.02522

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00005	0.33000	0.00004	2.50000	0.00005
	0.02000	0.02936	0.33000	0.02897	2.50000	0.02850
sg13g2_IOPadTriOut30mA	0.02000	-0.00004	0.33000	-0.00005	2.50000	-0.00004
	0.02000	0.02938	0.33000	0.02897	2.50000	0.02846
sg13g2_IOPadTriOut4mA	0.02000	0.00007	0.33000	0.00006	2.50000	0.00006
	0.02000	0.02938	0.33000	0.02897	2.50000	0.02845

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	-0.00005	0.33000	-0.00001	2.50000	-0.00002
	!c2p_en	0.02000	-0.02369	0.33000	-0.02467	2.50000	-0.02522
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00004	0.33000	0.00008	2.50000	0.00007
	!c2p_en	0.02000	-0.02369	0.33000	-0.02467	2.50000	-0.02522
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	-0.00007	0.33000	-0.00006	2.50000	-0.00006
	!c2p_en	0.02000	-0.02370	0.33000	-0.02468	2.50000	-0.02522

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00005	0.33000	0.00004	2.50000	0.00005
	!c2p_en	0.02000	0.02936	0.33000	0.02897	2.50000	0.02850
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	-0.00004	0.33000	-0.00005	2.50000	-0.00004
	!c2p_en	0.02000	0.02938	0.33000	0.02897	2.50000	0.02846
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00007	0.33000	0.00006	2.50000	0.00006
	!c2p_en	0.02000	0.02938	0.33000	0.02897	2.50000	0.02845